



4, 5, 6, 7, P

P 13
4 12
5 12
6 9
7 1

Name

Number.....

HSC Mathematics

Assessment Task 1 - 2016

November 2015

Time Allowed - 60 minutes +5 mins reading

Instructions: Calculators may be used in any parts of the task. For 1 Mark Questions, the correct answer is sufficient to receive full marks. For Questions worth more than 1 Mark, necessary working MUST be shown to receive full marks.

Multiple Choice	/4
Question 5	/10
Question 6	/9
Question 7	/12
Question 8	/12
Total	/47

Fill in the correct answer on the answer sheet - Questions 1 - 4 are worth 1 mark each

1. A parabola has a focus of $(-a, 0)$ and a directrix at $x = a$. Its equation is:

A $y^2 = 4ax$

B $y^2 = -4ax$

C $x^2 = 4ay$

D $x^2 = -4ay$

P

2. The derivative of $\frac{2}{x^2}$ is

A $-4x^{-3}$

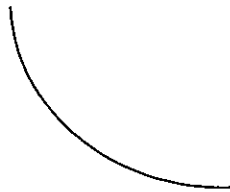
B $\frac{1}{x}$

C $\frac{-1}{x^3}$

D $\frac{-1}{2x^3}$

H 5

3. For the curve,



A $f'(x) > 0$ and $f''(x) < 0$

B $f'(x) > 0$ and $f''(x) > 0$

C $f'(x) < 0$ and $f''(x) < 0$

D $f'(x) < 0$ and $f''(x) > 0$

H 7

4. The gradient of the tangent to $y = 2x^3 - 3x - 5$ at $(-1, 2)$ is

A -2

B 3

C $\frac{1}{2}$

D $-\frac{1}{3}$

H 5

Question 5 (10 Marks) (Begin a new sheet of paper)

Marks

a) Differentiate i) $y = 4x^4 + 4x^3 - 7x - 2$

2

ii) $y = x\sqrt{x}$

2

b) Find the derivative in simplest factorised form of

i) $y = (2 - 3x)^7$

2

ii) $y = \frac{2x-1}{(x+1)^2}$

4

Question 6 (9 Marks) (Begin a new sheet of paper)

For the curve $y = 5x^3 - 3x^5$

i) Show that the x intercepts are $x = 0$ and $x = \pm 1.3$ to 1 decimal place.

1

ii) Find the coordinates of the stationary points.

2

iii) Determine the nature of each stationary point.

3

iv) Sketch the curve. (half page approximately)

3

Question 7 12 Marks (Begin a new sheet of paper)

- a) Prove that the locus of a point which is equidistant from the point $(-1, 2)$ and the line $y = -2$ is a parabola and find its equation. 3

- b) For the parabola $(y + 1)^2 = 4(x - 2)$:

- i) Find the coordinates of the vertex 1
- ii) Find the coordinates of the focus 1
- iii) Find the equation of the directrix 1
- iv) Sketch the parabola showing all these features. 2



- c) Find the equation of the locus of a point which is equidistant from $3x + 4y = 12$ and the y axis. 4

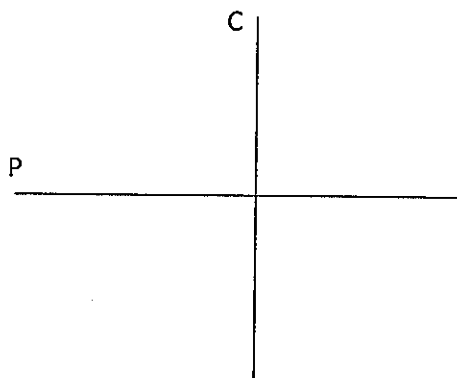
Question 8 (12 Marks) (Begin a new sheet of paper)

- a) A rectangular prism has edges x metres, $2x$ metres and y metres. The sum of all 12 edges is 36 metres.

- i) Show that $y = 9 - 3x$ 1
- ii) Show that its volume is given by $V = 18x^2 - 6x^3$. H 4 2
- iii) Hence find the maximum volume. 3

- b) Two people set out from towns on roads which meet at right angles and walk towards the intersection. Pat is 25 km from the intersection and walks at 4 km/h. Chris is 20 km from the intersection and walks at 3 km/h.

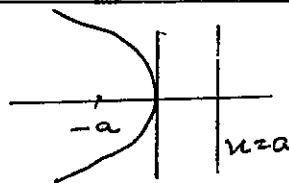
- i) Show that the distance D km apart after t hours is
 $D = \sqrt{25t^2 - 320t + 1025}$ H 4 2
- ii) Find the time in hours and minutes when Pat and Chris are closest. 3
- iii) Hence find their minimum distance apart. 1



END OF TEST

Solutions to 2016 Mathematics Task 1

1. B
 $y^2 = -4ax$



2. $y = 2x^{-2}$
 $y' = -4x^{-3}$ A

3. $f' < 0, f'' > 0$ D

4. $y' = 6x^2 - 3$
 $x = -1, m = 6 - 3 = 3$ B

Question 5

a) i) $y' = 12x^2 - 7$

ii) $y = x^{3/2}$
 $y' = \frac{3}{2} x^{1/2} = \frac{3\sqrt{x}}{2}$

b) i) $y' = 7(2-3x)^6 \cdot -3$
 $= -21(2-3x)^6$

ii) $y' = \frac{(x+1)^2 \cdot 2 - (2x-1) \cdot 2(x+1)}{(x+1)^4}$
 $= \frac{2(x+1)\{x+1-2x+1\}}{(x+1)^4}$
 $= \frac{2(2-x)}{(x+1)^3}$

Question 6

i) $y = 5x^3 - 3x^5$
 $y = 0 \quad x^3(5-3x^2) = 0$

$x = 0 \quad \vee \quad x^2 = \frac{5}{3}$

$x = 0 \quad \vee \quad x = \pm\sqrt{\frac{5}{3}} \doteq \pm 1.3$

ii) $y' = 15x^2 - 15x^4$
 $= 15x^2(1-x^2)$
 $= 15x^2(1-x)(1+x)$

Stationary points when

$x = 0, x = 1, x = -1$

$y = 0, y = 2, y = -2$

iii) Nature of (0,0)

x	0^-	0	0^+
y'	$+$	0	$+$

(0,0) is a horizontal point of inflexion.

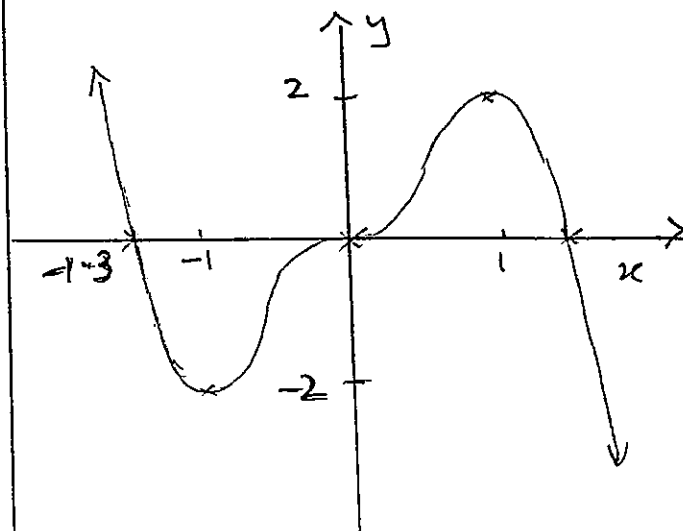
$y'' = 30x - 60x^3$

when $x = 1, y'' = -30$

Negative $\therefore (1, 2)$ is a maximum.

when $x = -1, y'' = 30$

positive $\therefore (-1, -2)$ is a minimum.



Question 7

a) $PS = PD$
 $PS^2 = PD^2$

$$(x+1)^2 + (y-2)^2 = (x-x)^2 + (y+2)^2$$

$$(x+1)^2 + y^2 - 4y + 4 = y^2 + 4y + 4$$

$$(x+1)^2 = 8y$$

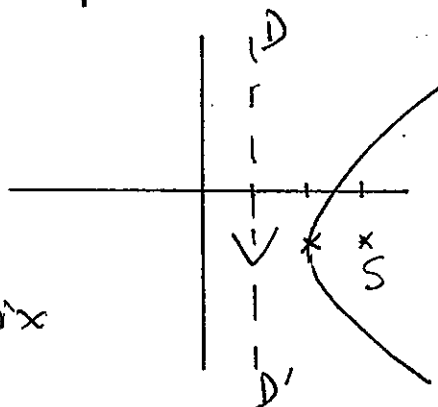
which is a parabola.

b)

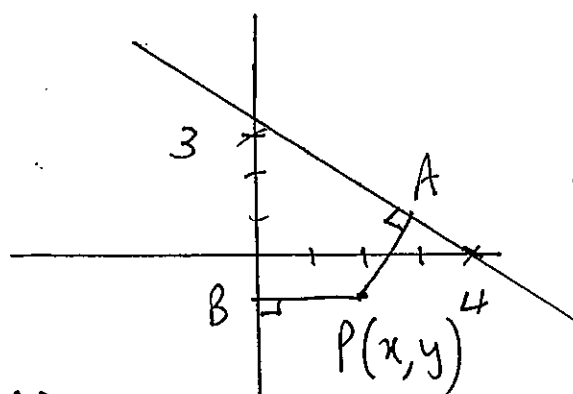
i) $V(2, -1)$

ii) $S(3, -1)$

iii) Directrix
 $x = 1$



c)



$$PA = PB$$

$$\left| \frac{3x+4y-12}{\sqrt{3^2+4^2}} \right| = |x|$$

$$3x+4y-12 = 5x$$

$$0 = 2x - 4y + 12$$

$$0 = x - 2y + 6 \quad \text{OR}$$

$$3x+4y-12 = -5x$$

$$8x+4y-12=0$$

$$2x+y-3=0$$

$\therefore$ locus is a pair of perpendicular lines

$$\begin{cases} x-2y+6=0 & + \\ 2x+y-3=0 \end{cases}$$

bisecting the angles of intersection.

Question 8

a) $4(x+2x+y) = 36$

$$3x+y=9$$

$$y = 9-3x$$

ii) $V = x \cdot 2x \cdot y$
 $= 2x^2(9-3x)$

$$V = 18x^2 - 6x^3$$

iii) For maximum volume $\frac{dV}{dx} = 0$

$$36x - 18x^2 = 0$$

$$18x(2-x) = 0$$

$$x = 0 \quad \text{or} \quad 2$$

$$\frac{d^2V}{dx^2} = 36 - 36x$$

$$\text{When } x = 2$$

$$\frac{d^2V}{dx^2} = 36 - 72$$

Negative $\therefore$ Maximum.

$$V = 18 \times 4 - 6 \times 8$$

$$V = 24 \text{ m}^3$$

8 b)

$$S = \frac{D}{T}$$

$$D = ST$$

In t hours

Pat travels $4t$ Km

$\therefore$ she is now $(25-4t)$ Km from intersection

In t hours

Chris travels $3t$ Km

she is now $(20-3t)$ Km from intersection

Distance apart is

$$D^2 = (25-4t)^2 + (20-3t)^2$$

$$= 625 - 200t + 16t^2 + 400 - 120t + 9t^2$$

$$D^2 = 25t^2 - 320t + 1025$$

$$D = \sqrt{25t^2 - 320t + 1025}$$

ii) For minimum distance, $\frac{dD^2}{dt} = 0$

$$\frac{d(D^2)}{dt} = 50t - 320$$

$$t = \frac{320}{50} = 6.4 \text{ hrs.}$$

$$= 6 \text{ hrs } 24 \text{ mins.}$$

$\frac{d^2 D^2}{dt^2} = 50t$ which is positive when t is positive. $\therefore D^2$ is minimum
ie D is minimum.

$$\text{iii) } D = \sqrt{25 \times 6.4^2 - 320 \times 6.4 + 1025}$$

$$= \sqrt{1} = 1$$

Minimum distance apart is 1 Km.

